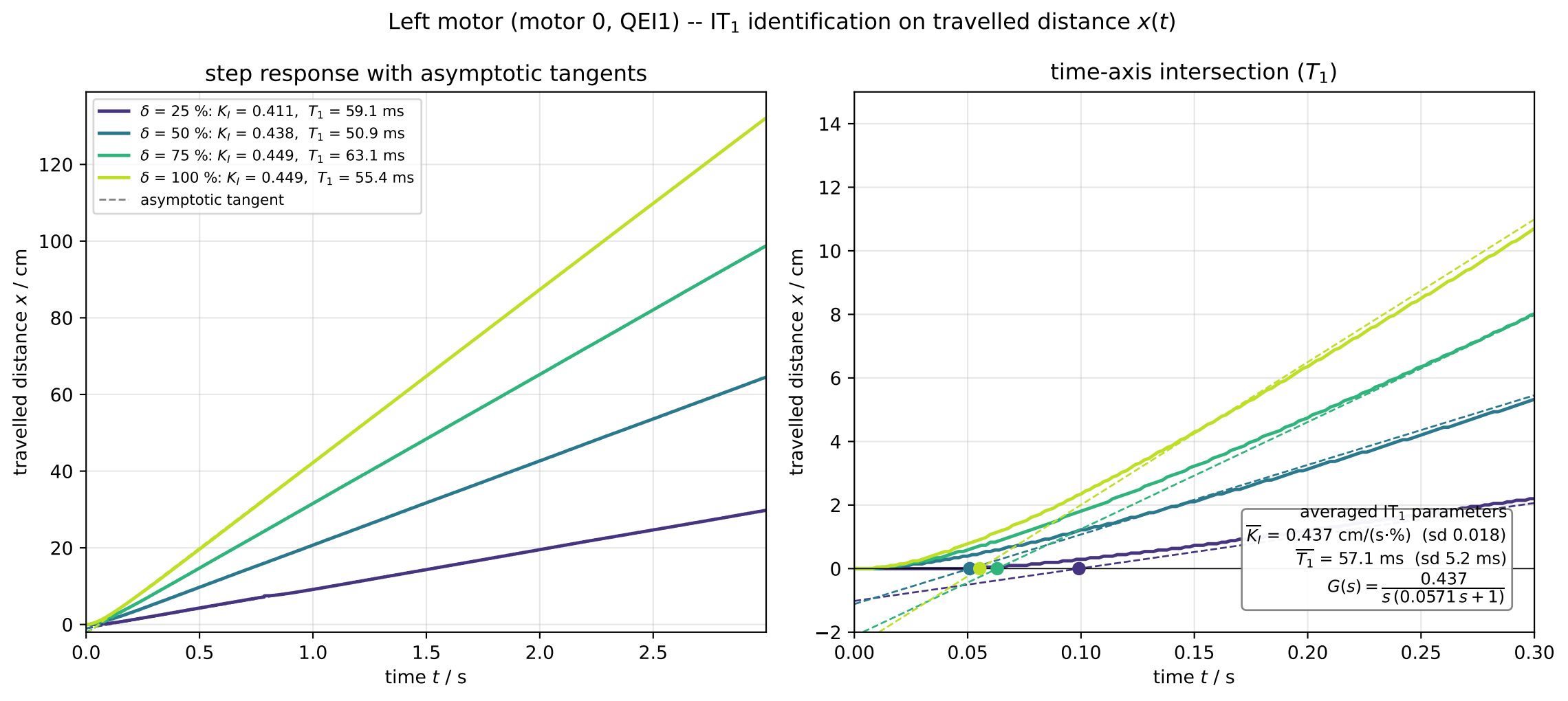
Left motor (motor 0, QEI1) -- IT$_1$ identification on travelled distance $x(t)$
step response with asymptotic tangents
δ = 25 %: K_I = 0.411, T_1 = 59.1 ms
δ = 50 %: K_I = 0.438, T_1 = 50.9 ms
δ = 75 %: K_I = 0.449, T_1 = 63.1 ms
δ = 100 %: K_I = 0.449, T_1 = 55.4 ms
asymptotic tangent
travelled distance x / cm
time t / s
time-axis intersection (T_1)
travelled distance x / cm
time t / s
averaged IT$_1$ parameters
$\overline{K_I}$ = 0.437 cm/(s·%) (sd 0.018)
$\overline{T_1}$ = 57.1 ms (sd 5.2 ms)
$G(s) = \frac{0.437}{s(0.0571\,s+1)}$